

Predictive Modeling (19 questions)

1. (Given a Dataset) Analyze this dataset and give me a model that can predict this response variable.

2. What could be some issues if the distribution of the test data is significantly different than the distribution of the training data?

- The model that has high training accuracy might have low test accuracy. Without further knowledge, it is hard to know which dataset represents the population data and thus the generalizability of the algorithm is hard to measure. This should be mitigated by repeated splitting of train vs test dataset (as in cross validation).
- When there is a change in data distribution, this is called the dataset shift. If the train and test data have a different distribution, then the classifier would likely overfit to the train data.
- This issue can be overcome by using a more general learning method.
- This can occur when:
 - $P(y|x)$ are the same but $P(x)$ are different. (covariate shift)
 - $P(y|x)$ are different. (concept shift)
- The causes can be:
 - Training samples are obtained in a biased way. (sample selection bias)
 - Train is different from test because of temporal, spatial changes. (non-stationary environments)
- Solution to covariate shift
 - importance weighted cv

3. What are some ways I can make my model more robust to outliers?

- We can have regularization such as L1 or L2 to reduce variance (increase bias).
- Changes to the algorithm:
 - Use tree-based methods instead of regression methods as they are more resistant to outliers. For statistical tests, use non-parametric tests instead of parametric ones.
 - Use robust error metrics such as MAE or Huber Loss instead of MSE.
- Changes to the data:
 - Winsorizing the data.

- Transforming the data (e.g., log).
- Remove them only if you're certain they're anomalies not worth predicting.

4. What are some differences you would expect in a model that minimizes squared error, versus a model that minimizes absolute error? In which cases would each error metric be appropriate?

- MSE is stricter to having outliers. MAE is more robust in that sense but is harder to fit the model for because it cannot be numerically optimized. So, when there are less variability in the model and the model is computationally easy to fit, we should use MAE, and if that's not the case, we should use MSE.
- MSE: easier to compute the gradient, MAE: linear programming needed to compute the gradient
- MAE more robust to outliers. If the consequences of large errors are great, use MSE
- MSE corresponds to maximizing likelihood of Gaussian random variables

5. What error metric would you use to evaluate how good a binary classifier is? What if the classes are imbalanced? What if there are more than 2 groups?

- Accuracy: proportion of instances you predict correctly. Pros: intuitive, easy to explain. Cons: works poorly when the class labels are imbalanced and the signal from the data is weak.
- AUROC: plot fpr on the x axis and tpr on the y axis for different threshold. Given a random positive instance and a random negative instance, the AUC is the probability that you can identify who's who. Pros: Works well when testing the ability of distinguishing the two classes. Cons: can't interpret predictions as probabilities (because AUC is determined by rankings), so can't explain the uncertainty of the model.
- logloss/deviance: Pros: error metric based on probabilities. Cons: very sensitive to false positives, negatives.
- When there are more than 2 groups, we can have k binary classifications and add them up for logloss. Some metrics like AUC is only applicable in the binary case.

6. What are various ways to predict a binary response variable? Can you compare two of them and tell me when one would be more appropriate? What's the difference between these? (SVM, Logistic Regression, Naive Bayes, Decision Tree, etc.)

- Things to look at: N, P, linearly separable?, features independent?, likely to overfit?, speed, performance, memory usage
- Logistic Regression
 - features roughly linear, problem roughly linearly separable;
 - robust to noise, use l1,l2 regularization for model selection, avoid overfitting;
 - the output comes as probabilities;
 - efficient and the computation can be distributed;
 - can be used as a baseline for other algorithms;
 - (-) can hardly handle categorical features.
- SVM
 - with a nonlinear kernel, can deal with problems that are not linearly separable;
 - (-) slow to train, for most industry scale applications, not efficient.
- Naive Bayes
 - computationally efficient when P is large by alleviating the curse of dimensionality;
 - works surprisingly well for some cases even if the condition doesn't hold;
 - with word frequencies as features, the independence assumption can be seen reasonable. So the algorithm can be used in text categorization;
 - (-) conditional independence of every other feature should be met.
- Tree Ensembles
 - good for large N and large P, can deal with categorical features very well;
 - non parametric, so no need to worry about outliers;
 - GBT's work better but the parameters are harder to tune;
 - RF works out of the box, but usually performs worse than GBT.
- Deep Learning
 - works well for some classification tasks (e.g., image);
 - used to squeeze something out of the problem.

7. What is regularization and where might it be helpful? What is an example of using regularization in a model?

- Regularization is useful for reducing variance in the model, meaning avoiding overfitting. For example, we can use L1 regularization in Lasso regression to penalize large coefficients.

8. Why might it be preferable to include fewer predictors over many?

- When we add irrelevant features, it increases model's tendency to overfit because those features introduce more noise. When two variables are correlated, they might be harder to interpret in case of regression, etc.
- curse of dimensionality
- adding random noise makes the model more complicated but useless
- computational cost
- ask someone for more details.

9. Given training data on tweets and their retweets, how would you predict the number of retweets of a given tweet after 7 days after only observing 2 days worth of data?

- Build a time series model with the training data with a seven-day cycle and then use that for a new data with only 2 days data.
- Ask someone for more details.
- Build a regression function to estimate the number of retweets as a function of time t .
- To determine if one regression function can be built, see if there are clusters in terms of the trends in the number of retweets.
- If not, we have to add features to the regression function.
- Features + # of retweets on the first and the second day -> predict the seventh day.
- https://en.wikipedia.org/wiki/Dynamic_time_warping

10. How could you collect and analyze data to use social media to predict the weather?

- We can collect social media data using twitter, Facebook, Instagram API's. Then, for example, for twitter, we can construct features from each tweet, e.g., the tweeted date, number of favorites, retweets, and of course, the features created from the tweeted content itself. Then use a multivariate time series model to predict the weather.
- Ask someone for more details.

11. How would you construct a feed to show relevant content for a site that involves user interactions with items?

- We can do so using building a recommendation engine. The easiest we can do is to show contents that are popular other users, which is still a valid strategy if for example the contents are news articles. To be more accurate, we can build a content based filtering or collaborative filtering. If there's enough user usage data, we can try collaborative filtering and recommend contents other similar users have consumed. If there isn't, we can recommend similar items based on vectorization of items (content-based filtering).

12. How would you design the people you may know feature on LinkedIn or Facebook?

- Find strong unconnected people in weighted connection graph
 - Define similarity as how strong the two people are connected
 - Given a certain feature, we can calculate the similarity based on
 - Friend connections (neighbors).
 - Check-in's people being at the same location all the time.
 - Same college, workplace.
 - Have randomly dropped graphs test the performance of the algorithm.
- News Feed Optimization
 - Affinity score: how close the content creator and the users are.
 - Weight: weight for the edge type (comment, like, tag, etc.). Emphasis on features the company wants to promote.
 - Time decay: the older the less important.

13. How would you predict who someone may want to send a Snapchat or Gmail to?

- For each user, assign a score of how likely someone would send an email to.
- The rest is feature engineering:
 - number of past emails, how many responses, the last time they exchanged an email, whether the last email ends with a question mark, features about the other users, etc.
- Ask someone for more details.
- People who someone sent emails the most in the past, conditioning on time decay.

14. How would you suggest to a franchise where to open a new store?

- Build a master dataset with local demographic information available for each location.
 - Local income levels, proximity to traffic, weather, population density, proximity to other businesses.
 - A reference dataset on local, regional, and national macroeconomic conditions (e.g., unemployment, inflation, prime interest rate, etc.).
 - Any data on the local franchise owner-operators, to the degree the manager.
- identify a set of KPIs acceptable to the management that had requested the analysis concerning the most desirable factors surrounding a franchise
 - Quarterly operating profit, ROI, EVA, pay-down rate, etc.
- Run econometric models to understand the relative significance of each variable.
- Run machine learning algorithms to predict the performance of each location candidate.

15. In a search engine, given partial data on what the user has typed, how would you predict the user's eventual search query?

- Based on the past frequencies of words shown up given a sequence of words, we can construct conditional probabilities of the set of next sequences of words that can show up (n-gram). The sequences with highest conditional probabilities can show up as top candidates.
- To further improve this algorithm,

- We can put more weight on past sequences which showed up more recently and near your location to account for trends.
- Show your recent searches given partial data.

16. Given a database of all previous alumni donations to your university, how would you predict which recent alumni are most likely to donate?

- Based on frequency and amount of donations, graduation year, major, etc., construct a supervised regression (or binary classification) algorithm.

17. Suppose you are Uber and you want to design a heatmap to recommend to drivers where to wait for a passenger. How would you approach this?

- Based on the past pickup location of passengers around the same time of the day, day of the week (month, year).
- Ask someone for more details.
- Based on the number of past pickups
 - account for periodicity (seasonal, monthly, weekly, daily, hourly).
 - special events (concerts, festivals, etc.) from tweets.

18. How would you build a model to predict a March Madness bracket?

- One vector each for team A and B. Take the difference of the two vectors and use that as an input to predict the probability that team A would win by training the model. Train the models using past tournament data and make a prediction for the new tournament by running the trained model for each round of the tournament
- Some extensions:
 - Experiment with different ways of consolidating the 2 team vectors into one (e.g., concatenating, averaging, etc.).
 - Consider using a RNN type model that looks at time series data.

19. You want to run a regression to predict the probability of a flight delay, but there are flights with delays of up to 12 hours that are really messing up your model. How can you address this?

Data Analysis (26 questions)

- 1. (Given a Dataset) Analyze this dataset and tell me what you can learn from it.**
- 2. What is R²? What are some other metrics that could be better than R² and why?**
- 3. What is the curse of dimensionality?**
- 4. Is more data always better?.**
- 5. What are advantages of plotting your data before performing analysis?**
- 6. How can you make sure that you don't analyze something that ends**
- 7. What is the role of trial and error in data analysis? What is the role of making a hypothesis before diving in?**
- 8. How can you determine which features are the most important in your model?**
- 9. How do you deal with some of your predictors being missing?**
- 10. You have several variables that are positively correlated with your response, and you think combining all of the variables could give you a good prediction of your response. However, you see that in the multiple linear regression, one of the weights on the predictors is negative. What could be the issue?**

11. Let's say you're given an unfeasible number of predictors in a predictive modeling task. What are some ways to make the prediction more feasible?

12. Now you have a feasible number of predictors, but you are sure that you don't need all of them. How would you perform feature selection on the dataset?

13. Your linear regression didn't run and communicates that there are an infinite number of best estimates for the regression coefficients. What could be wrong?

14. You run your regression on different subsets of your data, and find that in each subset, the beta value for a certain variable varies wildly. What could be the issue here?

15. What is the main idea behind ensemble learning? If I had many different models that predicted the same response variable, what might I want to do to incorporate all the models? Would you expect this to perform better than an individual model or worse?

16. How could you use GPS data from a car to determine the quality of a driver?

17. Given accelerometer, altitude, and fuel usage data from a car, how would you determine the optimum acceleration pattern to drive over hills?

18. Given position data of NBA players in a season's games, how would you evaluate a basketball player's defensive ability?

19. How would you quantify the influence of a Twitter user?

20. Given location data of golf balls in games, how would construct a model that can advise golfers where to aim?

21. You have 100 mathletes and 100 math problems. Each mathlete gets to choose 10 problems to solve. Given data on who got what problem correct, how would you rank the problems in terms of difficulty?

22. You have 5000 people that rank 10 kinds of sushi in terms of saltiness. How would you aggregate this data to estimate the true saltiness rank in each kind?

23. Given data on congressional bills and which congressional representatives co-sponsored the bills, how would you determine which other representatives are most similar to yours in voting behavior? How would you evaluate who is the most liberal? Most republican? Most bipartisan?

24. How would you come up with an algorithm to detect plagiarism in online content?

25. You have data on all purchases of customers at a grocery store. Describe to me how you would program an algorithm that would cluster the customers into groups. How would you determine the appropriate number of clusters to include?

26. Let's say you're building the recommended music engine at Spotify to recommend people music based on past listening history. How would you approach this problem?

Product Metrics (14 questions)

- 1. What would be good metrics of success for an advertising-driven consumer product? (Buzzfeed, YouTube, Google Search, etc.) A service-driven consumer product? (Uber, Flickr, Venmo, etc.).**
- 2. What would be good metrics of success for a productivity tool? (Evernote, Asana, Google Docs, etc.) A MOOC? (edX, Coursera, Udacity, etc.)**
- 3. What would be good metrics of success for an e-commerce product? (Etsy, Groupon, Birchbox, etc.) A subscription product? (Netflix, Birchbox, Hulu, etc.) Premium subscriptions? (OKCupid, LinkedIn, Spotify, etc.)**
- 4. What would be good metrics of success for a consumer product that relies heavily on engagement and interaction? (Snapchat, Pinterest, Facebook, etc.) A messaging product? (GroupMe, Hangouts, Snapchat, etc.)**
- 5. What would be good metrics of success for a product that offered in-app purchases? (Zynga, Angry Birds, other gaming apps)**
- 6. A certain metric is violating your expectations by going down or up more than you expect. How would you try to identify the cause of the change?**
- 7. Growth for total number of tweets sent has been slow this month. What data would you look at to determine the cause of the problem?**

8. You're a restaurant and are approached by Groupon to run a deal. What data would you ask from them in order to determine whether or not to do the deal?

9. You are tasked with improving the efficiency of a subway system. Where would you start?

10. Say you are working on Facebook News Feed. What would be some metrics that you think are important? How would you make the news each person gets more relevant?

11. How would you measure the impact that sponsored stories on Facebook News Feed have on user engagement? How would you determine the optimum balance between sponsored stories and organic content on a user's News Feed?

12. You are on the data science team at Uber, and you are asked to start thinking about surge pricing. What would be the objectives of such a product and how would you start looking into this?

13. Say that you are Netflix. How would you determine what original series you should invest in and create?

14. Let's say that you are scheduling content for a content provider on television. How would you determine the best times to schedule content?

Programming (14 questions)

1. Write a function to calculate all possible assignment vectors of $2n$ users, where n users are assigned to group 0 (control), and n users are assigned to group 1 (treatment).

2. Given a list of tweets, determine the top 10 most used hashtags.

3. Program an algorithm to find the best approximate solution to the knapsack problem¹ in a given time.

4. Program an algorithm to find the best approximate solution to the travelling salesman problem² in a given time.

5. You have a stream of data coming in of size n , but you don't know what n is ahead of time. Write an algorithm that will take a random sample of k elements. Can you write one that takes $O(k)$ space?

- https://en.wikipedia.org/wiki/Reservoir_sampling

6. Write an algorithm that can calculate the square root of a number.

- https://www.quora.com/What-is-the-method-to-calculate-a-square-root-by-hand?redirected_qid=664405
- https://en.wikipedia.org/wiki/Newton's_method#Square_root_of_a_number

7. Given a list of numbers, can you return the outliers?

8. When can parallelism make your algorithms run faster?

9. What are the different types of joins? What are the differences between them?

- (INNER) JOIN: Returns records that have matching values in both tables LEFT

- (OUTER) JOIN: Return all records from the left table, and the matched records from the right table
- RIGHT (OUTER) JOIN: Return all records from the right table, and the matched records from the left table
- FULL (OUTER) JOIN: Return all records when there is a match in either left or right table.

10. Why might a join on a subquery be slow? How might you speed it up?

- Change the subquery to a join.

11. Describe the difference between primary keys and foreign keys in a SQL database.

- Primary keys are columns whose value combinations must be unique in a specific table so that each row can be referenced uniquely. Foreign keys are columns that references columns (often primary keys) in other tables.

12. Given a COURSES table with columns course_id and course_name, a FACULTY table with columns faculty_id and faculty_name, and a COURSE_FACULTY table with columns faculty_id and course_id, how would you return a list of faculties who teach a course given the name of a course?

- `select faculty_name from faculty_id c join (select faculty_id from (select course_id from COURSES where course_name=xxx) as a join COURSE_FACULTY b on a.course_id = b.course_id) d on c.faculty_id = d.faculty_id.`

13. Given a IMPRESSIONS table with ad_id, click (an indicator that the ad was clicked), and date, write a SQL query that will tell me the click-through-rate of each ad by month.

- `select id, average(click) from (select count(click) as click from IMPRESSIONS group by id,month(date)) group by id.`

14. Write a query that returns the name of each department and a count of the number of employees in each:

EMPLOYEES containing: Emp_ID (Primary key) and Emp_Name

EMPLOYEE_DEPT containing: Emp_ID (Foreign key) and Dept_ID (Foreign key)

DEPTS containing: Dept_ID (Primary key) and Dept_Name.

- `select Dept_Name, count(1) from DEPTS a right join EMPLOYEE_DEPT b on a.Dept_id = b.Dept_id group by Dept_Name.`

Statistical Inference (16 questions)

1. In an A/B test, how can you check if assignment to the various buckets was truly random?
2. What might be the benefits of running an A/A test, where you have two buckets who are exposed to the exact same product?
3. What would be the hazards of letting users sneak a peek at the other bucket in an A/B test?
4. What would be some issues if blogs decide to cover one of your experimental groups?
5. How would you conduct an A/B test on an opt-in feature?
6. How would you run an A/B test for many variants, say 20 or more?
7. How would you run an A/B test if the observations are extremely right skewed?
 - lower the variability by modifying the KPI
 - cap values
 - percentile metrics
 - log transform
 - <https://www.quora.com/How-would-you-run-an-A-B-test-if-the-observations-are-extremely-right-skewed>

8. I have two different experiments that both change the sign-up button to my website. I want to test them at the same time. What kinds of things should I keep in mind?

9. What is a p-value? What is the difference between type-1 and type-2 error?

- A p-value is defined such that under the null hypothesis less than the fraction p of events have parameter values more extreme than the observed parameter. It is not the probability that the null hypothesis is wrong.
- type-1 error: rejecting H_0 when H_0 is true
- type-2 error: not rejecting H_0 when H_a is true

10. Suppose you are AirBnB and you want to test the hypothesis that a greater number of photographs increases the chances that a buyer selects the listing. How would you test this hypothesis?

11. How would you design an experiment to determine the impact of latency on user engagement?

12. What is maximum likelihood estimation? Could there be any case where it doesn't exist?

- A method for parameter optimization (fitting a model). We choose parameters so as to maximize the likelihood function (how likely the outcome would happen given the current data and our model).
- maximum likelihood estimation (MLE) is a method of [estimating](#) the [parameters](#) of a [statistical model](#) given observations, by finding the parameter values that maximize the [likelihood](#) of making the observations given the parameters. MLE can be seen as a special case of the [maximum a posteriori estimation](#) (MAP) that assumes a [uniform prior distribution](#) of the parameters, or as a variant of the MAP that ignores the prior and which therefore is [unregularized](#).
- for gaussian mixtures, non-parametric models, it doesn't exist

13. What's the difference between a MAP, MOM, MLE estimator? In which cases would you want to use each?

14. What is a confidence interval and how do you interpret it?

15. What is unbiasedness as a property of an estimator? Is this always a desirable property when performing inference? What about in data analysis or predictive modeling?

16. What is Selection Bias?

- Selection bias is a kind of error that occurs when the researcher decides who is going to be studied. It is usually associated with research where the selection of participants isn't random. It is sometimes referred to as the selection effect. It is the distortion of statistical analysis, resulting from the method of collecting samples. If the selection bias is not considered, then some conclusions of the study may not be accurate.
- The types of selection bias include:
 - Sampling bias: It is a systematic error due to a non-random sample of a population causing some members of the population to be less likely to be included than others resulting in a biased sample.
 - Time Interval bias: A trial may be terminated early at an extreme value (often for ethical reasons), but the extreme value is likely to be reached by the variable with the largest variance, even if all variables have a similar mean.
- Data: When specific subsets of data are chosen to support a conclusion or rejection of bad data on arbitrary grounds, instead of according to previously stated or generally agreed criteria.
- Attrition: Attrition bias is a kind of selection bias caused by attrition (loss of participants) discounting trial subjects/tests that did not run to completion.

Communication (5 questions)

1. Explain to me a technical concept related to the role that you're interviewing for.

2. Introduce me to something you're passionate about.

3. How would you explain an A/B test to an engineer with no statistics background? A linear regression?

- A/B testing, or more broadly, multivariate testing, is the testing of different elements of a user's experience to determine which variation helps the business achieve its goal more effectively (i.e. increasing conversions, etc..) This can be copy on a web site, button colors, different user interfaces, different email subject lines, calls to action, offers, etc.

4. How would you explain a confidence interval to an engineer with no statistics background? What does 95% confidence mean?

5. How would you explain to a group of senior executives why data is important?

20 Probability Interview Problems Asked By Top-Tech Companies & Wall Street

1. **[Facebook - Easy]** There is a fair coin (one side heads, one side tails) and an unfair coin (both sides tails). You pick one at random, flip it 5 times, and observe that it comes up as tails all five times. What is the chance that you are flipping the unfair coin?
2. **[Lyft - Easy]** You and your friend are playing a game. The two of you will continue to toss a coin until the sequence HH or TH shows up. If HH shows up first, you win. If TH shows up first, your friend wins. What is the probability of you winning?
3. **[Google - Easy]** What is the probability that a seven-game series goes to 7 games?
4. **[Facebook - Easy]** Facebook has a content team that labels pieces of content on the platform as spam or not spam. 90% of them are diligent raters and will label 20% of the content as spam and 80% as non-spam. The remaining 10% are non-diligent raters and will label 0% of the content as spam and 100% as non-spam. Assume the pieces of content are labeled independently from one another, for every rater. Given that a rater has labeled 4 pieces of content as good, what is the probability that they are a diligent rater?
5. **[Bloomberg - Easy]** Say you draw a circle and choose two chords at random. What is the probability that those chords will intersect?
6. **[Amazon - Easy]** 1/1000 people have a particular disease, and there is a test that is 98% correct if you have the disease. If you don't have the disease, there is a 1% error rate. If someone tests positive, what are the odds they have the disease?
7. **[Facebook - Easy]** There are 50 cards of 5 different colors. Each color has cards numbered between 1 to 10. You pick 2 cards at random. What is the probability that they are not of same color and also not of same number?
8. **[Tesla - Easy]** A fair six-sided die is rolled twice. What is the probability of getting 1 on the first roll and not getting 6 on the second roll?
9. **[Facebook - Easy]** What is the expected number of rolls needed to see all 6 sides of a fair die?
10. **[Microsoft - Easy]** Three friends in Seattle each told you it's rainy, and each person has a $\frac{1}{3}$ probability of lying. What is the probability that Seattle is rainy? Assume the probability of rain on any given day in Seattle is 0.25.
11. **[Uber - Easy]** Say you roll three dice, one by one. What is the probability that you obtain 3 numbers in a strictly increasing order?

12. **[Bloomberg - Medium]** Three ants are sitting at the corners of an equilateral triangle. Each ant randomly picks a direction and starts moving along the edge of the triangle. What is the probability that none of the ants collide? Now, what if it is k ants on all k corners of an equilateral polygon?
13. **[Two Sigma - Medium]** What is the expected number of coin flips needed to get two consecutive heads?
14. **[Amazon - Medium]** How many cards would you expect to draw from a standard deck before seeing the first ace?
15. **[Robinhood - Medium]** A and B are playing a game where A has $n+1$ coins, B has n coins, and they each flip all of their coins. What is the probability that A will have more heads than B?
16. **[Airbnb - Medium]** Say you are given an unfair coin, with an unknown bias towards heads or tails. How can you generate fair odds using this coin?
17. **[Quora - Medium]** Say you have N i.i.d. draws of a normal distribution with parameters μ and σ . What is the probability that k of those draws are larger than some value Y ?
18. **[Spotify - Hard]** A fair die is rolled n times. What is the probability that the largest number rolled is r , for each r in $1..6$?
19. **[Snapchat - Hard]** There are two groups of n users, A and B, and each user in A is friends with those in B and vice versa. Each user in A will randomly choose a user in B as their best friend and each user in B will randomly choose a user in A as their best friend. If two people have chosen each other, they are mutual best friends. What is the probability that there will be no mutual best friendships?
20. **[Tesla - Hard]** Suppose there is a new vehicle launch upcoming. Initial data suggests that any given day there is either a malfunction with some part of the vehicle or possibility of a crash, with probability p which then requires a replacement. Additionally, each vehicle that has been around for n days must be replaced. What is the long-term frequency of vehicle replacements?

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14. **[Amazon - Medium]** How many cards would you expect to draw from a standard deck before seeing the first ace?
15. **[Robinhood - Medium]** A and B are playing a game where A has $n+1$ coins, B has n coins, and they each flip all of their coins. What is the probability that A will have more heads than B?
16. **[Airbnb - Medium]** Say you are given an unfair coin, with an unknown bias towards heads or tails. How can you generate fair odds using this coin?

17. **[Quora - Medium]** Say you have N i.i.d. draws of a normal distribution with parameters μ and σ . What is the probability that k of those draws are larger than some value Y ?
18. **[Spotify - Hard]** A fair die is rolled n times. What is the probability that the largest number rolled is r , for each r in $1 \dots 6$?
19. **[Snapchat - Hard]** There are two groups of n users, A and B , and each user in A is friends with those in B and vice versa. Each user in A will randomly choose a user in B as their best friend and each user in B will randomly choose a user in A as their best friend. If two people have chosen each other, they are mutual best friends. What is the probability that there will be no mutual best friendships?
20. **[Tesla - Hard]** Suppose there is a new vehicle launch upcoming. Initial data suggests that any given day there is either a malfunction with some part of the vehicle or possibility of a crash, with probability p which then requires a replacement. Additionally, each vehicle that has been around for n days must be replaced. What is the long-term frequency of vehicle replacements?

20 Statistics Problems Asked By FANG & Hedge Funds

1. **[Facebook - Easy]** How would you explain a confidence interval to a non-technical audience?
2. **[Two Sigma - Easy]** Say you are running a multiple linear regression and believe there are several predictors that are correlated. How will the results of the regression be affected if they are indeed correlated? How would you deal with this problem?
3. **[Uber - Easy]** Describe p-values in layman's terms.
4. **[Facebook - Easy]** How would you build and test a metric to compare two user's ranked lists of movie/tv show preferences?
5. **[Microsoft - Easy]** Explain the statistical background behind power.
6. **[Twitter - Easy]** Describe A/B testing. What are some common pitfalls?

7. **[Google - Medium]** How would you derive a confidence interval from a series of coin tosses?
8. **[Stripe - Medium]** Say you model the lifetime for a set of customers using an exponential distribution with parameter λ , and you have the lifetime history (in months) of n customers. What is your best guess for λ ?
9. **[Lyft - Medium]** Derive the mean and variance of the uniform distribution $U(a, b)$.
10. **[Google - Medium]** Say we have $X \sim \text{Uniform}(0, 1)$ and $Y \sim \text{Uniform}(0, 1)$. What is the expected value of the minimum of X and Y ?
11. **[Spotify - Medium]** You sample from a uniform distribution $[0, d]$ n times. What is your best estimate of d ?
12. **[Quora - Medium]** You are drawing from a normally distributed random variable $X \sim N(0, 1)$ once a day. What is the approximate expected number of days until you get a value of more than 2?
13. **[Facebook - Medium]** Derive the expectation for a geometric distributed random variable.
14. **[Google - Medium]** A coin was flipped 1000 times, and 550 times it showed up heads. Do you think the coin is biased? Why or why not?
15. **[Robinhood - Medium]** Say you have n integers $1 \dots n$ and take a random permutation. For any integers i, j let a swap be defined as when the integer i is in the j th position, and vice versa. What is the expected value of the total number of swaps?
16. **[Uber - Hard]** What is the difference between MLE and MAP? Describe it mathematically.
17. **[Google - Hard]** Say you have two subsets of a dataset for which you know their means and standard deviations. How do you calculate the blended mean and standard deviation of the total dataset? Can you extend it to K subsets?
18. **[Lyft - Hard]** How do you randomly sample a point uniformly from a circle with radius 1?
19. **[Two Sigma - Hard]** Say you continually sample from some i.i.d. uniformly distributed $(0, 1)$ random variables until the sum of the variables exceeds 1. How many times do you expect to sample?
20. **[Uber - Hard]** Given a random Bernoulli trial generator, how do you return a value sampled from a normal distribution?

